

Automated Retinopathy Diagnosis System with Real-Time Image Processing



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Dedication

This project is dedicated to all the people who have lost their vision or their lives due to diabetic retinopathy, and to those who are still struggling with this disease every day.

May this humble effort contribute, even in a small way, to raising awareness, supporting early detection, and preventing blindness for countless others in the future.

Abstract

Diabetic Retinopathy (DR) is one of the primary causes of avoidable blindness globally, as a result of the increasing incidence of diabetes mellitus and the associated microvasculature changes in the retina. The lack of access to advanced ophthalmological services and the prohibitive cost of conventional imaging technologies in rural and marginalised communities make early detection of DR challenging. This research presents the design and development of an automated diagnostic platform that integrates deep learning to bridge this healthcare gap. The study involved the comparative analysis of three state-of-the-art Convolutional Neural Network (CNN) models: VGG16, ResNet50, and InceptionV3. These architectures were trained, validated and tested on the APTOS 2019 Blindness Detection dataset, using a stratified 70:15:15 train/validation/test data split in order to account for class imbalances. In order to maintain diagnostic accuracy, a series of preprocessing steps were undertaken, including pixel-wise normalization, color space conversion, and image resizing to 224x224 pixels. In addition, a comprehensive data analysis was performed, revealing 128 duplicate images and a mean blur score of 258.54 to avoid inflated performance scores. Our experiments revealed that InceptionV3 is the best-performing architecture for multi-class classification, achieving a test accuracy of 76.36% and a macro F1-score of 0.53. This is because it can extract features at multiple scales, enabling it to identify subtle features associated with pathology, such as microaneurysms and neovascularization, more accurately than the sequential VGG16 (73.64%) or the deeper ResNet50 (73.27%). The best-performing model was then incorporated in a cross-platform mobile app built with Flutter and TensorFlow Lite, with a secure Firebase-based cloud server for data storage. It includes secure login, live image capture via the smartphone camera, and automatic diagnostic reporting and guidance. This study concludes that the proposed system is a scalable, viable solution for early DR screening in resource-limited settings, which could decrease the worldwide burden of visual impairment.

Table of Contents

Chapter 1: Introduction.....	6
1.1 Challenges in Healthcare Accessibility.....	6
1.2 Role of Artificial Intelligence in Medical Imaging.....	6
1.3 Aims and Objectives.....	6
1.3.1 Primary Aim.....	6
1.3.2 Specific Objectives.....	7
1.4 Research Questions.....	7
Chapter 2: Literature review.....	8
2.1 Diabetic Retinopathy and the Need for Early Detection.....	8
2.2 Artificial Intelligence and Deep Learning in Medical Imaging.....	8
2.3 CNN-Based Approaches for Diabetic Retinopathy Detection.....	9
2.4 Emerging Technologies for Accessible Screening.....	9
2.5 Image Preprocessing and Data Augmentation Techniques.....	10
2.6 Datasets for Diabetic Retinopathy Detection.....	10
2.7 Evaluation Metrics for Model Performance.....	11
2.8 Research Gap.....	11
2.9 Summary of Literature Review.....	12
Chapter 3: Methodology.....	14
3.1 Research Design and Approach.....	14
3.2 Data Collection and Acquisition.....	15
3.3 Exploratory Data Analysis.....	15
3.3.1 Dataset Overview and Labeling.....	15
3.3.2 Qualitative Visual Inspection.....	15
3.3.3 Statistical and Color Distribution Analysis.....	15
3.3.4 Technical Quality and Structural Assessment.....	16
3.4 Data Preprocessing.....	16
3.4.1 Image Resizing.....	16
3.4.2 Color Space Standardization.....	16
3.4.3 Pixel Value Normalization.....	16
3.4.4 Dataset Splitting and Stratification.....	17

3.5 Model Development.....	17
3.5.1 Convolutional Neural Networks (CNNs).....	17
3.5.3 Model Training.....	18
3.5.4 Model Selection and Application Integration.....	18
3.6 System Evaluation and Performance Framework.....	18
3.6.1 Overview of Evaluation Strategy.....	18
3.6.2 Core Statistical Classification Metrics.....	18
3.7 System Implementation and Integration.....	19
3.7.1 Mobile Application Development.....	19
3.7.2 Cloud Infrastructure and Security Protocols.....	19
Chapter 4: Implementation.....	20
4.1 Importing Libraries.....	20
4.2 Dataset Extraction and Loading.....	20
4.3 Exploratory Data Analysis (EDA).....	21
4.3.1 Dataset Overview.....	21
4.3.2 Class Distribution Analysis.....	21
4.3.3 Image Dimension Analysis.....	22
4.3.4 Sample Image Visualization.....	22
4.3.5 Pixel Intensity Analysis.....	23
4.3.6 RGB Channel Distribution.....	23
4.3.7 Duplicate Image Detection.....	23
4.3.8 Blur Detection Analysis.....	24
4.3.9 Edge Detection Analysis.....	24
4.4 Data Preprocessing.....	25
4.4.1 Dataset Splitting.....	25
4.4.2 Directory Structure Creation.....	25
4.4.3 Image Copying Process.....	26
4.4.4 Image Data Generators.....	26
4.4.5 Data Loading.....	26
4.5 Model Development.....	27
4.5.1 Base Model Selection.....	27
4.5.2 Custom Model Architecture.....	28
4.5.3 Model Compilation.....	29
4.6 Model Training.....	29
4.6.1 Training Execution.....	29

4.6.2 Training Curves Function.....	29
4.7 Model Saving.....	30
4.8 Model Evaluation.....	30
Chapter 5: Results, Discussion and Evaluation.....	32
5.1 Exploratory Data Analysis (EDA) Findings.....	32
5.1.1 Dataset Distribution Analysis.....	32
5.1.2 Image Characteristics Analysis.....	33
5.1.3 Visual Inspection of Samples.....	33
5.1.4 Pixel Intensity & Color Distribution.....	33
5.1.5 Data Quality Assessment.....	34
5.2 Model Training Performance.....	35
5.2.1 VGG16 Training Performance.....	35
5.2.2 ResNet50 Training Performance.....	36
5.2.3 InceptionV3 Training Performance.....	36
5.3 Model Evaluation Results.....	37
5.3.1 VGG16 Evaluation.....	37
5.3.2 ResNet50 Evaluation.....	38
5.3.3 InceptionV3 Evaluation.....	38
5.4 Comparative Analysis of Models.....	39
5.4.1 Summary of Performance Metrics.....	40
5.4.2 Deep Dive into Architectural Performance.....	40
5.4.3 Minority Class Sensitivity Analysis.....	40
5.4.4 Computational Efficiency vs. Accuracy.....	40
5.5 Discussion of Results.....	41
5.5.1 Key Findings from EDA and Models.....	41
5.5.2 Model Strengths and Limitations.....	41
5.6 Summary of Results.....	41
Chapter 6: Critical Evaluation, Limitations and Conclusion.....	42
6.1 Critical Evaluation.....	42
6.2 Limitations.....	42
6.3 Conclusion.....	42
REFERENCES.....	44

List of Tables

Table 2.1 Summarized Literature Review.....	15
Table 5.1. Performance Metrics of Models.....	39

List of Figures

Figure 2.1 Normal Retina vs Diabetic Retinopathy. Source: Lions Eye Institute.....	10
Figure 2.2 Architecture of CNN. Source: GeeksforGeeks.....	11
Figure 3.1 Methodology flow (Self).....	17
Figure 4.1 Code for Analyzing Class Distribution.....	23
Figure 4.2 Code for Image Dimension Analysis.....	23
Figure 4.3 Code for Visualizing Sample Images.....	24
Figure 4.4 Code for Analyzing Pixel Distribution.....	24
Figure 4.5 Code for Analyzing RGB Channel Distribution.....	25
Figure 4.6 . Code for Duplicate Image Detection.....	25
Figure 4.7 Code for Blur Detection.....	26
Figure 4.8 Code for Analyzing Edge Detection.....	26
Figure 4.9 Custom Model Implementation.....	28
Figure 4.10 Code for Training Model.....	28
Figure 4.11 Code for Plotting Curves.....	29
Figure 4.12 Function for Model Evaluation.....	30
Figure 5.1. Class Distribution.....	31
Figure 5.2. Sample Images.....	32
Figure 5.3. Pixel Intensity Distribution.....	32
Figure 5.4. RGB Channel Distribution.....	33
Figure 5.5. Edge Detection.....	33
Figure 5.6. VGG16 Loss & Accuracy Curve.....	34
Figure 5.7. ResNet50 Loss & Accuracy Curve.....	35
Figure 5.8. InceptionV3 Loss & Accuracy Curve.....	35
Figure 5.9. VGG16 Evaluation Merics.....	36
Figure 5.10. VGG16 Confusion Matrix.....	36
Figure 5.11. Resnet50 Evaluation Merics.....	37
Figure 5.12. Resnet50 Confusion Matrix.....	37
Figure 5.13. InceptionV3 Evaluation Metrics.....	38
Figure 5.14. InceptionV3 Confusion Matrix.....	38
Figure 5.15 User Authentication Interface.....	40
Figure 5.16 Patient Enrollment and Data Entry Form.....	41
Figure 5.17 Image Selection Interface.....	41
Figure 5.18 Patient Management Dashboard.....	42
Figure 5.19 Automated Diagnostic Report with Severity Classification.....	43
Figure 5.20 Use Case Diagram (self).....	43
Figure 5.21 Sequence Diagram (self).....	44

Chapter 1 : Introduction

Diabetes mellitus is a global health challenge of the twenty first century with tremendous significance and high rate of growth. The rising rates of diabetes in developed and developing countries have led to the rising number of long term diseases, most of which have significant detrimental effects on the quality of life. Diabetic retinopathy (DR) is one of the most severe and vision threatening complications among them. Due to continuous effect on retinal microvasculature by high blood sugar levels, diabetic retinopathy gradually causes the deterioration of the retinal microvasculature and may eventually cause permanent blindness when untreated or untimely diagnosed. It is still one of the major causes of avoidable visual impairment in working-age adults in the global population. It is a well-established clinical evidence that a timely therapeutic intervention, which is diagnosed early in life, would greatly reduce the chances of severe vision loss. The success of early detection however, lies on availability and accessibility of systematic retinal screening programs.

1.1 Challenges in Healthcare Accessibility

Specialist eye care services are not distributed equally even though the ophthalmic imaging technologies have improved. Some low-resource and rural areas have a shortage of qualified ophthalmologists and lack of modern diagnostic devices. Traditional retinal cameras are usually expensive, immobile and normally found in specialised clinical settings. Moreover, the process of the interpretation of the retinal images traditionally involves the work of qualified clinical judges, which further limits the screening capacity. As a result, a significant percentage of diabetic patients is not screened or diagnosed with the disease at very late stages. This inequality shows a severe gap between the clinical need to screen at a young age and the realities of the healthcare systems that are present. This gap needs unique, scaled and cost effective solutions that can help in assisting early diagnosis without losing diagnostic accuracy.

1.2 Role of Artificial Intelligence in Medical Imaging

The past few years have seen breakthrough possibilities in medical image analysis due to advances in Data Science and artificial intelligence (AI). Deep learning, specifically the application of Convolutional Neural Networks (CNNs) have shown great capacity to extract complex visual features of high dimensional image data automatically. In the area of ophthalmology, CNN systems have demonstrated positive results in categorizing retinal fundus images based on the severity of diabetic retinopathy. In contrast to conventional machine learning methods which are based on hand-designed features, the deep learning structures permit end-to-end learning where the models discover subtle patterns of pathology in raw image inputs. Although previous studies have shown promising degrees of accuracy in the controlled experiment conditions, there are still unresolved questions on the generalisability of the results on varied datasets, computational cost-efficiency, and the possibility of implementation by non-high resource clinical systems.

1.3 Aims and Objectives

1.3.1 Primary Aim

The main purpose of the current study is to design and develop a low-cost, automated diagnostic platform that will integrate mobile technology, real-time image processing, and deep learning to support the early diagnosis of diabetic retinopathy. The aim of this project is to provide medical practitioners with the assistance they need to prevent under-resourced or remote communities (or even remote areas) from losing their sight without resorting to costly conventional imaging technology by comparing several CNN models (VGG16, ResNet50, and InceptionV3) and implementing the most efficient model.

1.3.2 Specific Objectives

To systematically achieve the primary aim, the study targets the following specific objectives:

- To develop a retinal image analysis system.
- To train and compare VGG16, ResNet50, and InceptionV3 based on accuracy and efficiency.
- To select the best model and deploy it in an Android application for real-time diagnosis.
- To evaluate system performance using accuracy, precision, recall, and sensitivity.

1.4 Research Questions

- How do VGG16, ResNet50, and InceptionV3 compare in their multi-class diagnostic accuracy?
- Which architecture is most effective at identifying Severe and Proliferative DR despite significant class imbalance?
- What are the trade-offs between architectural complexity and real-time processing speed for mobile deployment?
- How do standardized preprocessing and normalization techniques influence the training convergence of these specific models?
- Which model offers the optimal balance for integration into a secure, cloud-based mobile diagnosis system?

Chapter 2 : Literature review

2.1 Diabetic Retinopathy and the Need for Early Detection

Diabetic retinopathy (DR) is among the most frequently occurring health issues of diabetes mellitus and a chief cause of avoidable blindness in the world. This disease is caused by long term high blood sugar levels that destroy the retinal micro-vessels causing impactful loss of vision with time, unless treated. Recent scoping reviews indicate that the global prevalence of DR has reached between 27% and 40% in high-risk regions, primarily due to the chronic deterioration of retinal microvasculature which leads to permanent damage to the fovea and macula (Global Prevalence of Diabetic Retinopathy Over the Last Decade, 2025). The International Diabetes Federation reports that the prevalence rate of diabetes in the world keeps rising in an alarming rate, thus making the populations vulnerable to the effects of DR. Furthermore, projections suggest that by 2050, the global diabetes prevalence will reach 1.31 billion people, making automated screening a critical public health necessity (Joshi et al., 2025). Early screening and diagnosis are thus critical in the prevention of significant visual and physical loss and lessening the burden on future healthcare services. Early screening and diagnosis are thus critical in the prevention of significant visual and physical loss and lessening of future healthcare services.

DIABETIC RETINOPATHY

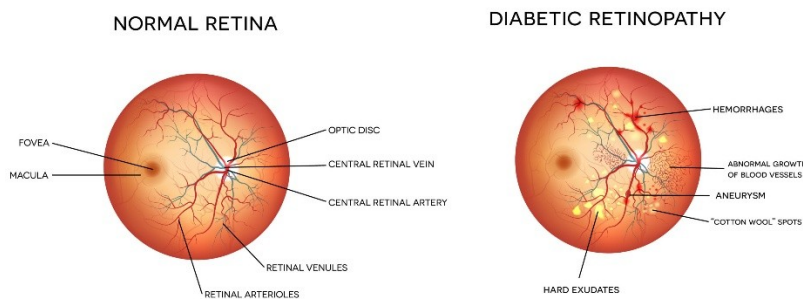


Figure : Literature review.1 Normal Retina vs Diabetic Retinopathy. Source: Lions Eye Institute

Most commonly used methods of screening diabetic retinopathy include a retinal fundi examination after which, trained ophthalmologists or retinal specialists typically examine these images manually. Despite the fact that this method may give reliable diagnosis, it has various limitations. In rural or underserved areas, access to specialist care is not usually possible, and the broad access to retinal imaging devices makes the expansion of screening programs even more challenging. This has led to a lot of patients being diagnosed at very advanced stages with some not being diagnosed at all. Such difficulties have stimulated researchers to consider automated and

technological-based ways to enhance accessibility and efficiency in the screening of diabetic retinopathy (Bellemo et al., 2019)

The recent advances in the field of artificial intelligence (AI) and medical image analysis have opened new possibilities in enhancing the process of diagnostics in ophthalmology. Specifically, deep learning methods have shown a great possibility of identifying pathological patterns in retinal images with high precision. AI-driven automated screening systems can support health care providers by detecting diseases as early as possible, decreasing the number of specialists who should work, and improving the process of diagnosis in great numbers.

2.2 Artificial Intelligence and Deep Learning in Medical Imaging

The use of artificial intelligence in healthcare has gained a significant role in the recent period, especially with respect to the analysis of medical images. Deep learning is a sub-discipline of machine learning that employs neural network structures to learn complicated patterns on large data volumes. One of these architectures, Convolutional Neural Networks (CNNs), have been found to be especially useful to image classification problems because the model has the capability of identifying spatial hierarchies and visual features in images.

CNN oriented models can extract features using raw images without the need to do manual feature engineering. This has enabled them to be extremely applicable in the medical imaging practices like tumor television, radiological analysis and ophthalmic illness diagnosis. Pratt et al. (2016) have shown that CNN models can be used successfully to classify retinal fundus images into various diabetic retinopathy score sets with a high level of accuracy, which shows that deep learning could be used in screening systems.

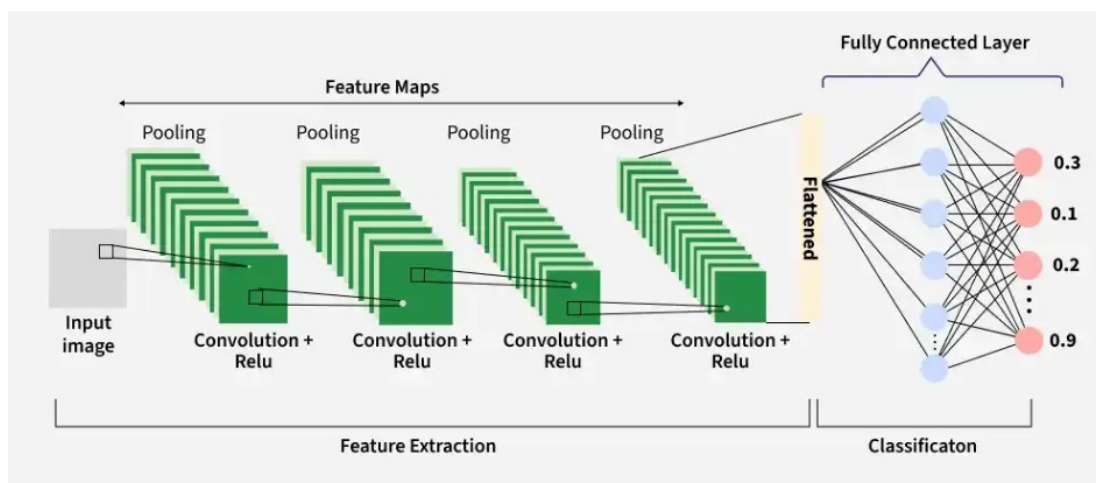


Figure : Literature review.2 Architecture of CNN. Source: GeeksforGeeks

The efficacy of using deep learning in ophthalmology has been further confirmed through massive research. Another significant research by Gulshan et al. (2016) introduced a deep learning algorithm that can identify diabetic retinopathy using retinal fundus images and perform similarly to board-certified ophthalmologists. Their research included training the model using a large sample of retinal images provided and labeled as well as testing its capabilities in various clinical

environments. The findings indicated that deep learning systems were capable of high sensitivity and specificity, thus could be applicable in the real-world screening.

In like manner, a multi-ethnic study was done by Ting et al, (2017), to test deep learning algorithms hoping to identify diabetic retinopathy and other eye disorders. Their results indicated that AI-based models could obtain the same level of results on a wide range of data as a professional clinician. The paper further highlighted that training processes and data should be of high quality and that extensive training sets are necessary to have confidence in the dependability and external validity of the findings in clinical settings.

2.3 CNN-Based Approaches for Diabetic Retinopathy Detection

An increasing literature on the use of CNN structures in detecting diabetic retinopathy has been examined. In designing neural networks, the early research was mainly concentrated towards classifying the retinal images into varying levels of severity of the disease. Ghosh et al. (2017) created a CNN-based system that recognizes and classifies the level of diabetic retinopathy experiments through retinal images. Their system showed promising accuracy and brought the aspect that the deep learning models have the potential to pick out subtle pathological features which might be hard to identify using the classical image processing methods.

One more contribution to the sphere was made by Quellec et al. (2017), who suggested a technique called deep image mining to screen diabetic retinopathy. In the method, neural networks are used to automatically detect useful features in retinal images without annotating disease features manually. Through automated feature extraction methods, the model also managed to enhance the accuracy of classification and minimized the need of handcrafting features engineering.

In addition to the development of the algorithm, studies have also been conducted into the applicability of automated screening systems. Gulshan et al. (2019) also performed additional tests of AI-based detectors of diabetic retinopathy in several diversity healthcare facilities worldwide. Their investigation showed that automated systems were capable of sustaining high diagnostic performance in various populations, which points to the possibility of using AI tools in clinical practices. Another emphasis that the authors noted is the need to develop user-friendly systems that can be easily adapted by the healthcare providers.

Equally, another system presented by Antonelli et al. (2019) is EyeArt, an automated retinal image decipher system intended to screen diabetic retinopathy at large scale. Tests conducted in clinical settings have shown that the system was highly diagnostic with a major reduction in the workload of the ophthalmologists. These results affirm the increasing evidence that AI tools can be useful aid tools in the habitual screening medical programs.

2.4 Transfer Learning in Diabetic Retinopathy Detection

Transfer learning has emerged as a highly effective approach in medical image analysis, particularly in scenarios where labeled datasets are limited. Instead of training deep neural networks from scratch, pretrained models such as VGG16, ResNet50, and InceptionV3 are fine-tuned on domain-specific datasets, significantly reducing training time and improving performance. Studies have shown that deep residual networks can achieve superior performance by addressing vanishing gradient problems (He et al., 2016), while very deep architectures such as

VGG are capable of learning hierarchical image features effectively (Simonyan & Zisserman, 2015). Additionally, the Inception architecture improves computational efficiency while maintaining high accuracy (Szegedy et al., 2016). In the context of diabetic retinopathy, deep learning models trained using transfer learning have demonstrated performance comparable to ophthalmologists in detecting disease from retinal images (Gulshan et al., 2016). Furthermore, similar approaches in medical imaging tasks have achieved expert-level classification performance, highlighting the robustness of transfer learning in healthcare applications (Esteva et al., 2017).

2.5 Emerging Technologies for Accessible Screening

Along with the enhancement of diagnostic algorithms, the recent researches have discussed the technologies, which allow an increased availability of automated screening systems. Cloud computing has become a significant part of massive medical image analysis. Lee et al. (2019) examined the applicability of the cloud-based services on diabetic retinopathy screenings and showed that by using secure cloud frameworks, one can process substantial amounts of retina scans and allow remote clinics to retrieve diagnostic outcomes. In such systems, it is a possibility to implement AI-enabled tools of screening over geographically distributed healthcare systems.

The accessibility of hardware has also been a research subject in this field. Akram et al. (2021) reviewed the viability of smartphone-retinal images of retinal imaging with low-cost lens caps. They concluded that mobile devices fitted with inexpensive lenses can take a retinal image of high quality to be analyzed automatically. The solution provided by this approach has a perspective in increasing the number of community screening services to underserved groups that might need the screening equipment provided the conventional retinal imaging system

2.6 Image Preprocessing and Data Augmentation Techniques

Preprocessing is significant in the medical image analysis using deep learning, enhancing the performance of the used models and making sure that the input images are appropriate to train neural networks. Some of the problems of the retinal fundus images include uneven illumination, noise, low contrast and change in imaging conditions. These pitfalls may adversely impact the capacity of convolutional neural networks to discover meaningful information in the information. Therefore, various preprocessing methods have been suggested to increase the quality of the image and normalize data prior to training.

Rahim et al. (2020) explored the various preprocessing methods used to detect diabetic retinopathy and established that image processing techniques like contrast enhancement, noise filtering, and image normalization were useful in enhancing the classification. These techniques are used to bring out significant retinal images including blood vessels, microaneurysms, and bleeding which are major signs of diabetic retinopathy. Other investigations, also, have proposed the use of histogram equalization and adaptive contrast improvement algorithms to enhance the visibility of retina pathology in retinal images.

The other useful method in deep learning applications is data augmentation. Medical data are typically small as clinical images need labeling by experts and thus they are not always collected correctly. Data augmentation is used to overcome this limitation by artificially enriching a training

dataset with geometric transformations of image rotation, scaling, flipping and cropping. Pires et al. (2020) showed that the robustness of CNN models and their generalization capacity could be significantly enhanced by using augmentation methods in the course of training diabetic retinopathy. The exposure of the neural network to variations of the same images helps the model to identify patterns in unseen data.

2.7 Datasets for Diabetic Retinopathy Detection

Another important aspect of the creation of the deep learning systems to detect diabetic retinopathy is the availability of high-quality datasets. Lots of labeled training data are needed to train CNN models to learn complicated visual patterns, which are related to disease development. Various publicly accessible retinal image databases have been commonly adopted by researchers to create and test automated screening algorithms.

A famous dataset that has been used by many researchers is the Kaggle Diabetic Retinopathy Detection dataset, which is composed of thousands of annotated retinal fundus images of the various stages of the disease. That dataset has been published as an international machine learning competition and has since served as a standard benchmark in the assessment of deep learning models in the diabetic retinopathy classification tasks.

The other common dataset is the EyePACS which is a vast number of retinal pictures taken in the course of routine clinical examination. This data was used by Gulshan et al. (2016) to train their deep learning algorithm and detect diabetic retinopathy automatically. The dataset offered a wide diversity of the images of various populations and imaging conditions and contributed to the better generalization ability of the trained model.

Moreover, the Messidor dataset has been widely applied to test diabetic retinopathy detectors, as well. Decenciere et al. (2014) have provided a research tool in automated retinal image analysis through the introduction of the Messidor database. The data consists of high-resolution retinal fundus photos with detailed annotations by the medical specialists. Due to its trustworthiness and standardized labeling, Messidor has been utilized extensively as a benchmark dataset in a number of studies that analyze deep learning methods.

It is frequently advised to use a number of datasets in the development and evaluation of the model to guarantee that deep learning models achieve consistency in various imaging setting and groups of people. The practice enhances credibility of the automated screening systems and minimizes the risk of biasing the model predictions.

2.8 Evaluation Metrics for Model Performance

Automated diabetic retinopathy detection systems should be evaluated using the proper metrics of performance to guarantee effective diagnosis. In medical image classification, it is not enough to evaluate just overall accuracy, models should also be able to detect diseased and non-diseased cases. Accuracy is the count of the rightly categorized images in the total set of samples, and it does not always make sense to use this measure alone, especially in the case of uneven datasets when the healthy pictures are in many times larger numbers than the diseased ones.

In order to give a more detailed examination, other metrics are often applied by the researchers, including precision, recall (sensitivity), specificity, and F1-score. The precision is a measure of the proportion of the correctly remembered positive cases whereas the recall is a measure of the sensitivity of the model to recognize the true cases of diseases. The F1-score is a metric that balances precision and recall providing a combined performance measure. The other commonly used measure is the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) that measures the effectiveness of the model to differentiate between the classes at the various levels of the decision. Other than studies like by Gulshan et al. (2016) indicate high values of AUC, which indicates the high diagnostic potential of deep learning models in the detection of diabetic retinopathy.

2.9 Research Gap

Despite its achieved success in the development of deep learning models to detect diabetic retinopathy, there are multiple limitations in the existing literature. Most of the current AI-based screening systems have been developed and tested within well-equipped healthcare facilities and hospitals with high resources. Such systems can be based on costly retinal imaging systems, high-end computer infrastructure and specialized support.

Comparatively, however, much less research has been conducted on the development of accessible screening solutions capable of functioning in a low-resource setting. In rural areas and developing nations especially, ophthalmology services are often inaccessible hence, it is hard to apply the traditional screening programs. Although recent studies have involved application of smartphone-based retinal imaging application and cloud-based diagnostic applications, they are yet to be extensively used or become common healthcare practices.

Moreover, high classification accuracy is usually one of the concerns that many studies aim at without regard to computational efficiency, deployment, and system usability. In real life healthcare practice, automated screening systems need to strike the balance between diagnostic performance and operational efficiency so that they can be both scalable and cost effective.

Consequently, the gap in knowledge is that more studies are required that combine the deep learning models, the accessible imaging technologies, and scaled implementation models to develop effective diabetic retinopathy screening systems applicable to underserved groups. To fill this gap, the study aims to create an automated diabetic retinopathy detection framework which will utilize CNN-based image classification with available imaging and deployment technologies.

2.10 Summary of Literature Review

Research Area	Key Researchers/Authors	Main Findings & Contributions
Need for Automated Screening	Bellemo et al. (2019)	Highlighted the limitations of manual screening in rural areas and the potential for AI-driven automated systems to improve

		accessibility and diagnostic efficiency.
Deep Learning Efficacy	Pratt et al. (2016); Gulshan et al. (2016); Ting et al. (2017)	Demonstrated that CNNs can classify DR severity with high precision, often matching the performance of board-certified ophthalmologists across diverse datasets.
CNN Methodologies	Ghosh et al. (2017); Quellec et al. (2017)	Developed systems capable of identifying subtle pathological features using automated feature extraction ("deep image mining") to enhance classification accuracy.
Clinical Implementation	Gulshan et al. (2019); Antonelli et al. (2019)	Confirmed the ability of AI tools like EyeArt to maintain high diagnostic performance in real-world clinical settings while reducing specialist workloads.
Accessible Technologies	Lee et al. (2019); Akram et al. (2021)	Explored cloud-based diagnostic retrieval for remote clinics and the use of low-cost smartphone-based retinal imaging for community screening.
Preprocessing & Augmentation	Rahim et al. (2020); Pires et al. (2020)	Identified that contrast enhancement and noise filtering significantly improve classification, while data augmentation increases model robustness against small clinical datasets.
Standardized Datasets	Decenciere et al. (2014); Kaggle/EyePACS	Established public databases like APTOS, EyePACS, and Messidor as essential benchmarks for training and evaluating DR detection models.
Evaluation Metrics	Gulshan et al. (2016)	Emphasized the necessity of utilizing metrics such as sensitivity (recall), specificity, and AUC-ROC to ensure reliable medical diagnosis beyond simple accuracy.
Identified Research Gap	Dissertant Analysis	Noted that most systems require high-end infrastructure; a critical gap exists for computationally efficient, low-resource mobile screening solutions for underserved groups

Chapter 3 : Methodology

The chapter provides the methodology that was used to develop an automated diabetic retinopathy (DR) diagnosis system, which in this case is based on the fact that the specialised ophthalmic services are lacking in the low-resource settings. The study design is applied and experimental, which combines the available mobile technology and the state-of-the-art deep learning models to design a portable, real-time screening device. In an effort to achieve maximized diagnostic accuracy and efficiency, this paper will comparatively analyze three pre-trained Convolutional Neural Network (CNN) models: VGG16, ResNet50 and InceptionV3. The best performing single model will then be chosen to be integrated into a secure, cloud-based mobile application. The subjects that follow describe the main research objectives, the overall design of the experiment, and the step-wise process of organizing, executing, and assessing the diagnostic system.

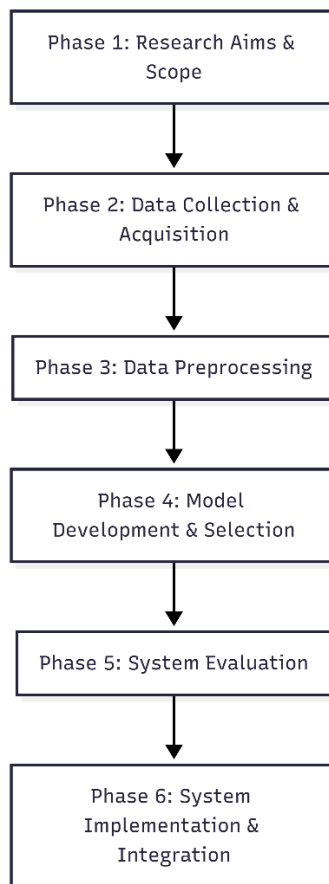


Figure : Methodology.3 Methodology flow (Self)

3.1 Research Design and Approach

It involves an applied research design and experimental research that draws on the development, comparative study, and approval of a practical technological prototype. The project is organized to use the Software Development Life Cycle (SDLC) to successfully manage liaisons between machine learning algorithms and mobile application development; the Waterfall methodology is used. It is a systematic method that divides the research into phases: Requirements and Planning, Design, Implementation, Testing, Deployment, and Maintenance. The main element of this experimental design is the comparative study of deep learning models. As opposed to the use of one architecture, the study compares VGG16, ResNet50 and InceptionV3. The study uses secondary data to develop the model by relying on APTOS 2019 Blindness Detection dataset on Kaggle which is highly validated and classifies images as No DR, Mild, Moderate, Severe, and Proliferative to form the five stages of severity associated with the visual impairment of the eye. Training and testing of the models will be done to identify the architecture that has the best balance between diagnostic accuracy and computational efficiency that can be deployed to mobile devices.

3.2 Data Collection and Acquisition

The foundation of the deep learning model training relies on robust and diverse secondary data. The primary dataset utilized for training, validating, and testing the chosen Convolutional Neural Network (CNN) architectures is the open-source APTOS 2019 Blindness Detection dataset, sourced from Kaggle. This dataset is widely cited in academic research and provides a vast collection of retinal fundus images. To facilitate accurate disease staging, the dataset categorizes these images into five clinically relevant severity levels: No Diabetic Retinopathy, Mild Diabetic Retinopathy, Moderate Diabetic Retinopathy, Severe Diabetic Retinopathy, and Proliferative Diabetic Retinopathy. The images within this dataset inherently exhibit variations in resolution, lighting, and visual clarity, which accurately reflects real-world clinical screening conditions and promotes the generalization of the diagnostic models.

3.3 Exploratory Data Analysis

Exploratory Data Analysis was employed as a pre-processing step to assess the nature of the dataset and make sure it is appropriate to train deep learning.

3.3.1 Dataset Overview and Labeling

The dataset was initially processed to identify the amount of data and the distribution of the data in the five stages of clinical diabetic retinopathy. The generation of a set of pathnames and labeling of each of the categories were automated using a script. This was a crucial process to help determine the possible imbalances between classes, which may require particular splitting or evaluation plans.

3.3.2 Qualitative Visual Inspection

A visual verification of the labeling process was performed to control the integrity of the labeling process with the help of the Matplotlib and OpenCV libraries. It was used to visualize randomly chosen samples of each directory to compare clinical pathological correlates of microaneurysms and hemorrhages with the digital images. It was an error checking method to ensure that pictures were properly classified in their respective folders.

3.3.3 Statistical and Color Distribution Analysis

To determine numeric attributes of the data, a set of analytic tools were used:

a) Image Dimension Analysis: Samples of all the classes were read to identify characteristics in width and height. The average output was then used to define a standardized input size of the CNN architectures.

b) Pixel Intensity Distribution: The analysis of brightness and contrast variances through the dataset was performed with the help of histogram plotting.

c) RGB Channel Analysis: Color channels were broken down and individually analyzed using OpenCV. This gave an understanding of colors in fundus images and the informative usefulness of certain channels in detecting lesions.

3.3.4 Technical Quality and Structural Assessment

To ensure the quality of the training pipeline, three quality control checks were carried out:

a) Duplicate Detection: MD5 hashing algorithm was applied in order to scan all the image files. The system determined the redundant data that might lead to leakage of data between the training and testing set by comparing the resulting hashes.

b) Blur Detection: Laplacian variance method was used to measure the image clarity. The sharpness of the images was measured using this method so that the models could be able to extract the fine-grained retinal features reliably.

c) Edge Detection Analysis: Canny edge detector was used to highlight the vascular tree and anatomical boundaries in the fundus scans. This procedure ensured that the morphological features to be used in the classification were not lost in the raw data.

3.4 Data Preprocessing

Data preprocessing is one of the more important stages in the creation of a strong deep learning model, especially when dealing with medical images where consistency and data quality are the direct correlates of predictive performance. The preprocessing methods were applied systematically in this study to make sure that the retinal images were uniform, significant, and fit well into a good training and testing mode.

3.4.1 Image Resizing

Retinal images obtained via various sources tend to have large variations in spatial dimension and resolution owing to the differences in imaging equipment and the imaging conditions. To mitigate this fluctuation and match it with the convolutional neural network structure, each image was resized to a constant spatial resolution size of 224 x 224 pixels. This standardization allows the model to read the inputs using a standard format, and retain the much-needed structural and pathological characteristics necessary to the correct classification.

3.4.2 Color Space Standardization

Digital photographs can be stored in various color channel configurations, as per the processing conditions. In order to preserve a consistency and proper interpretation of visual features, the images were all standardized into a standard representation of color. This measure will make sure

that the color channel configuration meets the expectations of the deep learning model and avoids distortions in the feature extraction and enhances the predictability of learned patterns.

3.4.3 Pixel Value Normalization

In order to increase the numerical stability throughout the training, pixel intensity values were brought to a normalized set of values. In particular, the initial intensity values, which generally vary between 0 and 255, were brought to the floating-point range of 0 to 1. By means of this normalization, the convergence can be achieved much faster, the risk of gradient-related problems is minimized, and the optimization process can be made more efficient. The training, validation, and testing dataset were normalized consistently, and this was also done at the inference stage, and the model lifecycle was consistent throughout.

3.4.4 Dataset Splitting and Stratification

In order to facilitate the training, validation and objective assessment of the model, the dataset was divided into three separate sets namely training, validation and testing. The strategy used in the splitting process was a stratified sampling strategy in order to maintain the proportional distribution of all five classes in each subset. Such a strategy helps to address the problem of a class imbalance and make sure that the model is trained and tested on representative samples of both categories. The dataset was partitioned into 70: 15: 15 proportion of training, validation and testing respectively.

3.5 Model Development

3.5.1 Convolutional Neural Networks (CNNs)

The theoretical framework of the image classification tasks used in this study is the deep learning, in particular, the application of Convolutional Neural Networks (CNNs). The CNNs are uniquely adapted to complicated medical images since they automatically acquire and extract hierarchical features using several hidden layers and filters. This automated capability of extracting features avoids feature engineering by humans, and substantially enhances the system capability of detecting delicate anatomy-related features that are linked to diabetic retinopathy, like fine lesions or blood vessel alterations.

3.5.2 Pre-Trained Architectures and Comparative Analysis

In order to achieve the highest diagnostic accuracy and maximize the performance in the resource-constrained mobile health system, it will be subjected to a strict comparative analysis of three most popular pre-trained CNN architectures, namely VGG16, ResNet50, and InceptionV3. Such models will be tested and compared in a systematic way based on their classification accuracy, performance efficiency and their overall aptitude to be deployed on mobile devices.

a) VGG16: A widely recognized deep CNN architecture known for its simplicity and uniform use of small convolution filters, which effectively capture detailed spatial features in images.

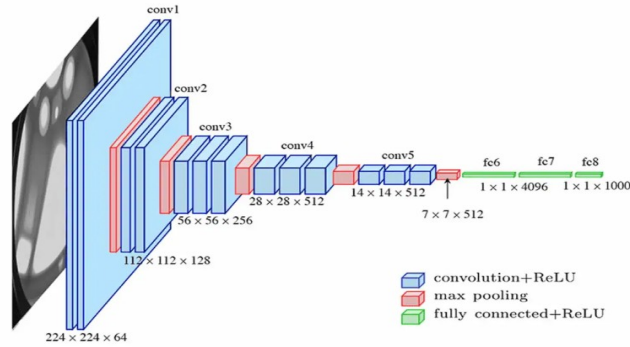


Figure : Methodology.4 VGG16 Architecture. Source: Viso Suite

b) ResNet50: A Residual Network architecture specifically designed to overcome the vanishing gradient problem often encountered in deep neural networks. By utilizing "skip connections," ResNet50 allows the network to be exceptionally deep without losing learning accuracy, making it highly effective for detecting minute changes in retinal scans.

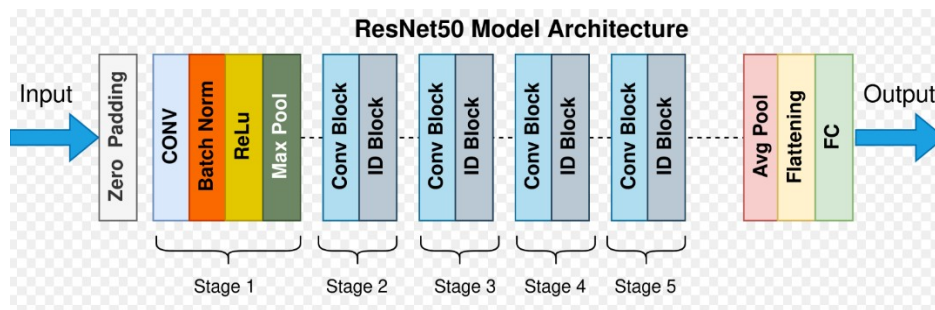


Figure : Methodology.5 ResNet50 Architecture. Source: Medium

c) InceptionV3: An advanced architecture that captures both local and global retinal features efficiently while optimizing computational resources, providing a strong baseline for comparison against the other models.

Inception V3

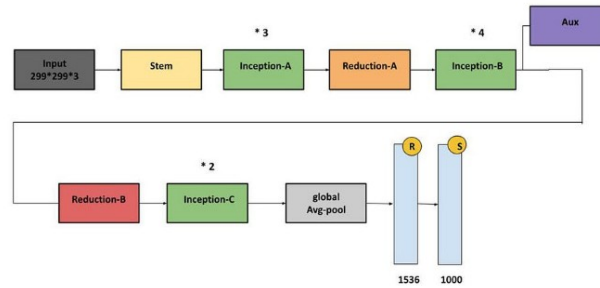


Figure : Methodology.6 InceptionV3 Architecture. Source: Medium

3.5.3 Model Training

The three chosen models will be implemented, trained as well as validated in Python programming language, along with the help of the TensorFlow as well as Keras deep learning frameworks. The models will be assembled in the course of the training phase with the Adam optimizer with the learning rate of 1e-5. The loss function will be categorical cross-entropy, which will be effectively used to tackle the multi-classification of diabetic retinopathy severity levels that exist in five levels. The models will be trained within 30 epochs, using the beforehand fixed training and validation subsets to update the weights of features and thoroughly observe the overfitting.

3.5.4 Model Selection and Application Integration

After the overall comparative analysis of VGG16, ResNet50, and InceptionV3, the most successful model will be chosen to be put into practice. In order to be able to integrate mobile in real time this best model will then be hosted on a secure cloud server. The system is able to speedily deliver captured retinal images to the server by deploying only the most accurate and efficient model to the cloud, process the inferences and provide viable diagnostic results in real time to healthcare professionals without overloading the local processing hardware of the mobile device.

3.6 System Evaluation and Performance Framework

3.6.1 Overview of Evaluation Strategy

To rigorously evaluate the diagnostic potential of the suggested Convolutional Neural Networks, the strong framework of evaluation was developed. After training and validation, the predictive power of the chosen models is evaluated on an entirely unknown test set. This guarantees the evaluation is appropriate to represent the system in the real-world in terms of generalisability of the system and clinical reliability of the system in detecting diabetic retinopathy without the system overfitting to the training data.

3.6.2 Core Statistical Classification Metrics

The output of the deep learning models is measured quantitatively with respect to a set of standard statistical classification metrics. Since medical screening devices need precision and thoroughness,

the assessment of various metrics is essential to guarantee a full picture of the trustworthiness of the system. The exact measures established in this work are:

- **Accuracy:** This measures the overall proportion of correctly predicted retinal images across all five severity classes (No DR, Mild, Moderate, Severe, Proliferative).
- **Precision:** This evaluates the exactness of the model, indicating the percentage of positive predictions made by the system that are genuinely correct.
- **Recall (Sensitivity):** A highly critical metric in medical diagnostics, recall measures the model's ability to correctly identify all actual positive cases of diabetic retinopathy, ensuring that diseased patients are not missed during screening.
- **F1-Score:** Serving as the harmonic mean of precision and recall, the F1-Score provides a balanced single measure of the model's performance. This is particularly useful for medical datasets where class distribution may be uneven.
- **Confusion Matrix:** A table comparing actual and predicted classes, showing correct predictions (diagonal) and misclassifications (off-diagonal), and helping analyze errors and compute other evaluation metrics.

3.7 System Implementation and Integration

3.7.1 Mobile Application Development

In a bid to make the automated diagnostic system accessible in remote locations, a specific mobile application was created as the main user interface through the Android Studio. This portable mobile application offers an easy, interactive user-experience, enabling healthcare professionals to use the camera on their phones to take retinal photos, upload them with ease and then achieve real-time diagnosis of these photos directly on their screens.

3.7.2 Cloud Infrastructure and Security Protocols

The system combines Firebase cloud services to support the safe and efficient management of the operations on the back-end and to take off-load of the local mobile device with the intense computational tasks. Firebase Firestore is the database that is used to store and retrieve the data on patients such as their names, ages, genders, and proposed diagnostic outcomes in real-time. Due to the sensitivity of the medical information being processed in the system, high levels of privacy are built within this particular cloud architecture. This involves the Firebase Authentication, when only authorized individuals may log in, and HTTPS encryption as well as two-factor authentication (2FA) to confirm that all information transfer will be safe and meet the general healthcare privacy standards.

Chapter 4 : Implementation

This chapter describes the practical implementation of the proposed system. Overall, it demonstrates how the proposed methodology is transformed into a working pipeline.

4.1 Exploratory Data Analysis (EDA)

Exploratory Data Analysis is carried out to get the sense of the characteristics of the data set prior to training a model. It gives details of the image distribution, quality and the structure of the dataset.

4.1.1 Class Distribution Analysis

The computation of the number of images in a given class is done by going through the class directories. A bar plot is subsequently used to visualize this data in order to examine the balance of dataset balance at the various levels of diabetic retinopathy severity. It is crucial to know the class distribution in order to detect any imbalance that can influence model training performance.

```

class_counts = {}

for cls in classes:
    path = os.path.join(data_dir, cls)
    class_counts[cls] = len(os.listdir(path))

# Plot
plt.figure(figsize=(8,5))
sns.barplot(x=list(class_counts.keys()), y=list(class_counts.values()))
plt.title("Class Distribution")
plt.xticks(rotation=45)
plt.show()

print(class_counts)

```

Figure : Implementation.7 Code for Analyzing Class Distribution

4.1.2 Image Dimension Analysis

The information about the image resolution is obtained by reading sample images of each class. The images width and height are gathered and averaged to comprehend the data resolution features of the entire dataset. This aids in determining an appropriate input image size to use in deep learning models, which makes them consistent throughout training.

```

widths = []
heights = []

for cls in classes:
    path = os.path.join(data_dir, cls)
    for img_name in os.listdir(path)[:100]:
        img_path = os.path.join(path, img_name)
        img = Image.open(img_path)
        w, h = img.size
        widths.append(w)
        heights.append(h)

print("Average Width:", np.mean(widths))
print("Average Height:", np.mean(heights))

```

Figure : Implementation.8 Code for Image Dimension Analysis

4.1.3 Sample Image Visualization

Each class has random images which are shown using Matplotlib. This is done to visually examine the dataset and learn the differences in retinal images in various disease stages. It also assists in checking the correctness of the datasets and also makes sure that the images have been loaded correctly out of their corresponding directories.

```

plt.figure(figsize=(12,8))
for i, cls in enumerate(classes):
    img_path = os.path.join(data_dir, cls, random.choice(os.listdir(os.path.join(data_dir, cls))))
    img = cv2.imread(img_path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    plt.subplot(2,3,i+1)
    plt.imshow(img)
    plt.title(cls)
    plt.axis('off')

plt.tight_layout()
plt.show()

```

Figure : Implementation.9 Code for Visualizing Sample Images

4.1.4 Pixel Intensity Analysis

Histogram plotting is used to analyze pixel intensity distribution. The picture is made flat and intensity value is plotted to comprehend the changes of brightness and contrast of images in the retina.

```

img_path = os.path.join(data_dir, classes[0], random.choice(os.listdir(os.path.join(data_dir, classes[0]))))
img = cv2.imread(img_path)

plt.figure(figsize=(10,4))
plt.hist(img.ravel(), bins=256)
plt.title("Pixel Intensity Distribution")
plt.show()

```

Figure : Implementation.10 Code for Analyzing Pixel Distribution

4.2.5 RGB Channel Distribution

The distributions of color channels (Red, Green, Blue) are discussed individually with the help of OpenCV. Each channel is created into histograms to gain insight into color composition and changes in fundus images.

```

img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

colors = ('r', 'g', 'b')

plt.figure(figsize=(10,4))
for i, col in enumerate(colors):
    hist = cv2.calcHist([img], [i], None, [256], [0,256])
    plt.plot(hist, color=col)

plt.title("RGB Channel Distribution")
plt.show()

```

Figure : Implementation.11 Code for Analyzing RGB Channel Distribution

4.2.6 Duplicate Image Detection

MD5 hashing is used to detect a duplicate image. The picture files are hashed and the number of repeated hashes is counted to get the redundant pictures in the dataset.

```
def get_hash(img_path):
    with open(img_path, 'rb') as f:
        return hashlib.md5(f.read()).hexdigest()

hashes = set()
duplicates = 0

for cls in classes:
    path = os.path.join(data_dir, cls)
    for img_name in os.listdir(path):
        img_path = os.path.join(path, img_name)
        h = get_hash(img_path)

        if h in hashes:
            duplicates += 1
        else:
            hashes.add(h)

print("Duplicate Images Found:", duplicates)
```

Figure : Implementation.12 . Code for Duplicate Image Detection

4.2.7 Blur Detection Analysis

Laplacian variance is used to determine the quality of an image. Low variance is a sign of a blurred image and the greater the values the sharper the image. The technique is applied to approximate the clarity of the images in the dataset.

```
def detect_blur(image_path):
    img = cv2.imread(image_path, 0)
    return cv2.Laplacian(img, cv2.CV_64F).var()

blur_scores = []

for cls in classes:
    path = os.path.join(data_dir, cls)
    for img_name in os.listdir(path)[:50]:
        score = detect_blur(os.path.join(path, img_name))
        blur_scores.append(score)

print("Average Blur Score:", np.mean(blur_scores))
```

Figure : Implementation.13 Code for Blur Detection

4.2.8 Edge Detection Analysis

The Canny edge detection algorithm is used to detect the edges. This is used to extract structural characteristics of retinal images and boundaries in fundus scans. It gives us an understanding of the anatomical features like blood vessels and retinal edges.

```

img = cv2.imread(img_path, 0)

edges = cv2.Canny(img, 100, 200)

plt.figure(figsize=(8,4))
plt.subplot(1,2,1)
plt.imshow(img, cmap='gray')
plt.title("Original")

plt.subplot(1,2,2)
plt.imshow(edges, cmap='gray')
plt.title("Edges")

plt.show()

```

Figure : Implementation.14 Code for Analyzing Edge Detection

4.3 Data Preprocessing

Preprocessing of the data is done to prepare the data to feed deep learning models. This involves splitting of datasets, arranging directories and normalization of images so that the input has a standard format.

4.3.1 Dataset Splitting

Stratified sampling is used to split the dataset into training, validation, and testing sets so that there is consistency in distribution of classes across the subsets. The split ratio is given as:

- 70% Training data
- 15% Validation data
- 15% Testing data

This makes sure that all classes are represented in each subset, enhancing the model generalization potential.

4.3.2 Data Loading

The datasets are loaded using the `flow_from_directory()` function. Images are automatically loaded in their corresponding directories, resized to 224×224 pixels and organized into batches to load and process them efficiently. This role also codes the class labels into categorical form that can be used in multi-class classification. The data is then readied and directly fed into deep learning models to be trained and evaluated.

4.4 Model Development

4.4.1 Base Model Selection

The architectures of pretrained convolutional neural networks are being used in extracting features on retinal images. The idea of transfer learning is utilized by the application of well-trained models that were trained on the ImageNet dataset.

To compare the architectures, three architectures are chosen:

- VGG16

- ResNet50
- InceptionV3

They are applied with their top classification layers disabled (`include_top=False`) in order to be customized to the diabetic retinopathy classification task. The learned visual features which include edges, textures, and patterns are leveraged through the use of the pretrained weights.

4.4.2 Custom Model Architecture

Each pretrained base model is additionally classified with a custom classification head. During the first step, all the layers of the base model are frozen to maintain the learned weights and avoid retraining convolutional filters. The proposed architecture starts with a Global Average Pooling layer to decrease the spatial dimensions, and then with a Batch Normalization layer to enhance the stability of training. This is then linked to a fully connected Dense layer of 256 neurons that have ReLU activation. A Dropout layer of 0.5 is used to minimize the chance of overfitting. Lastly, the network is completed with a Dense output layer with Softmax activation to achieve multi-classification. In general, this design allows to extract features efficiently out of pretrained models and adapts them to the needs of the particular dataset.

```
def build_model(base_model):

    # Freeze base model layers
    for layer in base_model.layers:
        layer.trainable = False

    x = base_model.output
    x = layers.GlobalAveragePooling2D()(x)
    x = layers.BatchNormalization()(x)
    x = layers.Dense(256, activation='relu')(x)
    x = layers.Dropout(0.5)(x)
    output = layers.Dense(NUM_CLASSES, activation='softmax')(x)

    model = models.Model(inputs=base_model.input, outputs=output)

    return model

vgg_model = build_model(vgg_base)
resnet_model = build_model(resnet_base)
inception_model = build_model(inception_base)
```

Figure : Implementation.15 Custom Model Implementation

4.5 Model Training

4.5.1 Training Execution

The models are trained using the prepared training dataset with validation data used for monitoring performance during training. All models are trained over 30 epochs, which is enough to allow them to learn features and converge. Training of VGG16, ResNet50 and InceptionV3 is performed individually to allow performance comparison. Training is done in batches by the use of the data generators that are generated in preprocessing.

```
history_vgg = vgg_model.fit(
    train_data,
    validation_data=val_data,
    epochs=30
)
```

Figure : Implementation.16 Code for Training Model

4.5.2 Training Curves Function

A custom function is implemented which is used to plot both the accuracy and loss at the epochs and visualize the training progress. It generates two main plots: training accuracy versus validation accuracy, and training loss versus validation loss. These visuals allow one to have a clear picture of how the model learned as the process of training continued. They are more effective in diagnosing possible problems like overfitting or underfitting to allow improved assessment and optimization of the model performance.

```
def plot_training_curves(history, model_name):

    # Accuracy
    plt.figure(figsize=(12,5))

    plt.subplot(1,2,1)
    plt.plot(history.history['accuracy'], label='Training Accuracy')
    plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
    plt.title(f'{model_name} - Accuracy Curve')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.legend()

    # Loss
    plt.subplot(1,2,2)
    plt.plot(history.history['loss'], label='Training Loss')
    plt.plot(history.history['val_loss'], label='Validation Loss')
    plt.title(f'{model_name} - Loss Curve')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.legend()

    plt.tight_layout()
    plt.show()
```

Figure : Implementation.17 Code for Plotting Curves

4.6 Model Evaluation

A dedicated evaluation function is implemented to assess the performance of each model on unseen test data. This function systematically generates predictions on the test dataset, converts the predicted probabilities into class labels using the argmax function, and compares them with the true labels. Based on this comparison, multiple evaluation metrics are computed to ensure a consistent

and structured analysis across all models. This approach allows reliable measurement of how well the trained models generalize to new data.

Model performance is evaluated using standard classification metrics, including accuracy, precision, recall, and F1-score. A detailed classification report is generated to provide class-wise performance insights, particularly important for distinguishing between different stages of diabetic retinopathy. In addition, a confusion matrix is created and visualized using a heatmap to present a clear breakdown of correct and incorrect predictions. This helps in identifying misclassification patterns between classes, which is critical for improving model reliability in medical diagnosis tasks.

```
def evaluate_model(model, test_data, model_name):

    print(f"\n Evaluating {model_name}...\n")

    # Predictions
    y_pred_probs = model.predict(test_data)
    y_pred = np.argmax(y_pred_probs, axis=1)
    y_true = test_data.classes

    class_names = list(test_data.class_indices.keys())

    # Accuracy
    acc = accuracy_score(y_true, y_pred)
    print(f"{model_name} Accuracy: {acc:.4f}")

    # Classification Report
    print("\nClassification Report:\n")
    print(classification_report(y_true, y_pred, target_names=class_names))

    # Confusion Matrix
    cm = confusion_matrix(y_true, y_pred)

    plt.figure(figsize=(7,5))
    sns.heatmap(cm, annot=True, fmt='d',
                xticklabels=class_names,
                yticklabels=class_names)

    plt.title(f"{model_name} - Confusion Matrix")
    plt.xlabel("Predicted")
    plt.ylabel("Actual")
    plt.show()

    return acc
```

Figure : Implementation.18 Function for Model Evaluation

Chapter 5 : Results, Discussion and Evaluation

This chapter presents the experimental results and evaluation of the trained models. It includes analysis of dataset characteristics, model performance comparison using evaluation metrics, and

interpretation of results. The chapter concludes with a discussion on the best-performing model and its clinical relevance.

5.1 Exploratory Data Analysis (EDA) Findings

The initial phase of the study involved a deep dive into the dataset to identify patterns, anomalies, and statistical properties that could influence model performance.

5.1.1 Dataset Distribution Analysis

The dataset comprises a total of 3,662 retinal fundus images categorized into five distinct severity levels. The distribution across these categories is as follows:

- **No DR:** 1,805 images
- **Mild:** 370 images
- **Moderate:** 999 images
- **Severe:** 193 images
- **Proliferative DR:** 295 images

A high class imbalance exists in the data set. The data of the "No DR" category constitutes almost half of the total data, and the most underrepresented category is the "Severe" one because it constitutes only about 5.2% of the total samples. Such an imbalance requires very close consideration with metrics that go beyond the accuracy measure, including F1-score and confusion matrices, so that the model does not get biased towards the majority class.

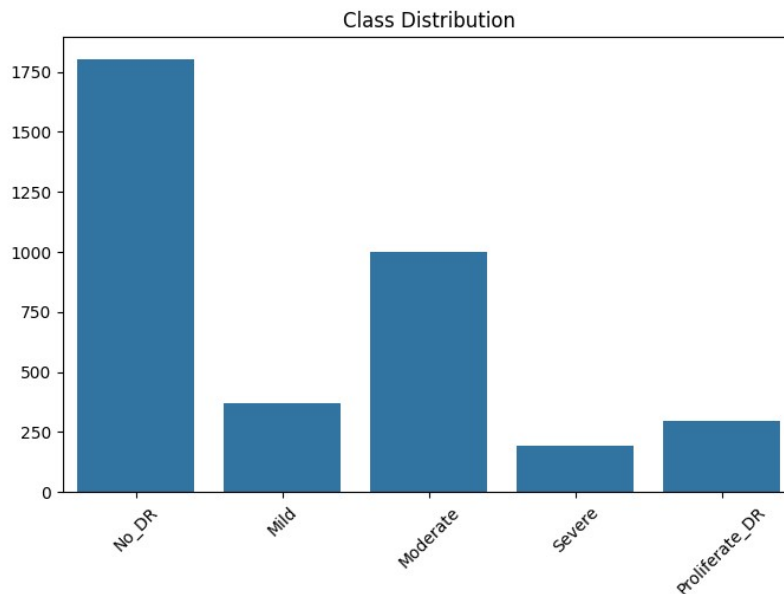


Figure : Results, Discussion and Evaluation.19. Class Distribution

5.1.2 Image Characteristics Analysis

To maintain consistency in the pipeline, a statistical analysis of the physical dimensions of the images was carried out. The analysis showed that the average width and height was 224.0 pixels.

The original data had different resolutions, but the homogeneous average would indicate a strongly standardized data after acquisition or a standard resizing approach was used in the early phase of loading the original data to ensure an aspect ratio of 1:1.

5.1.3 Visual Inspection of Samples

A representative picture of each of the five classes was studied in order to be able to correlate the clinical symptoms with the digital information.

- **No DR:** obeyed clear retinal surfaces with distinct blood vessels and no lesions.
- **Mild to Moderate:** The occurrence of microaneurysms and initial hemorrhages were observed.
- **Severe and Proliferative DR:** These images showed more advanced disease characteristics such as hard exudates (yellowish spots), large cotton wool spots and neovascularization (development of new abnormal blood vessels).

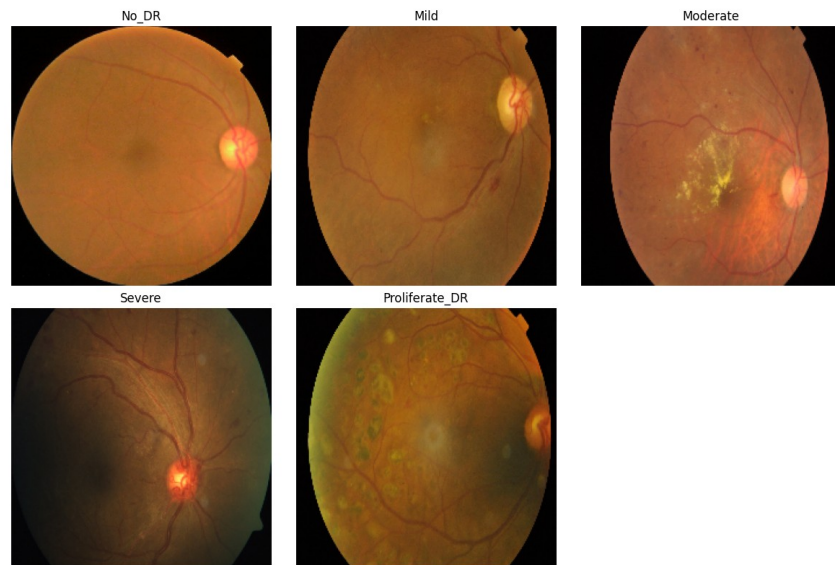


Figure : Results, Discussion and Evaluation.20. Sample Images

5.1.4 Pixel Intensity & Color Distribution

The histogram of the pixel intensities showed that the distribution is multimodal. The steep spike at the 0-intensity point is the black circular edges of fundus photography.

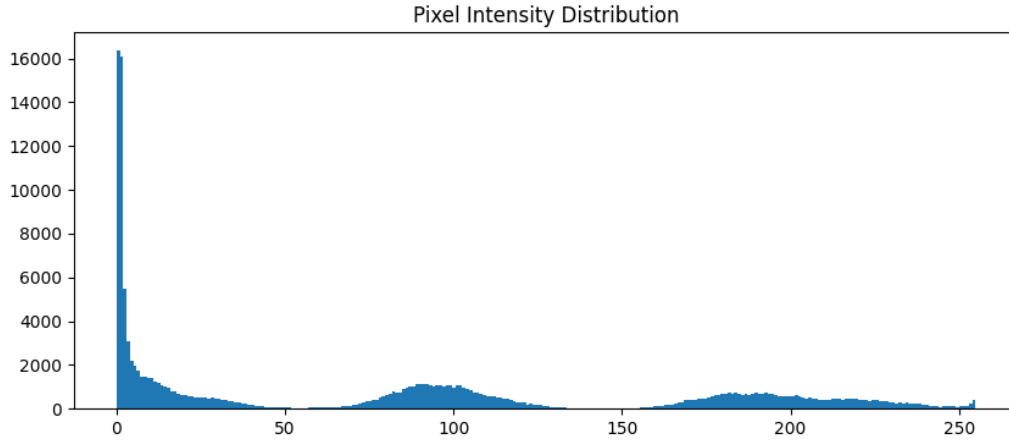


Figure : Results, Discussion and Evaluation.21. Pixel Intensity Distribution

The Red channel had more intense peaks in the high frequencies (150-250), which is in line with the reddish-orange colour of the human eye. The distribution in the Green channel was more central as it is frequently mentioned in literature as the most informative channel to detect retinal lesions. The Blue channel was held together at the lower intensity levels and it added least contribution to the structure detail of the images.

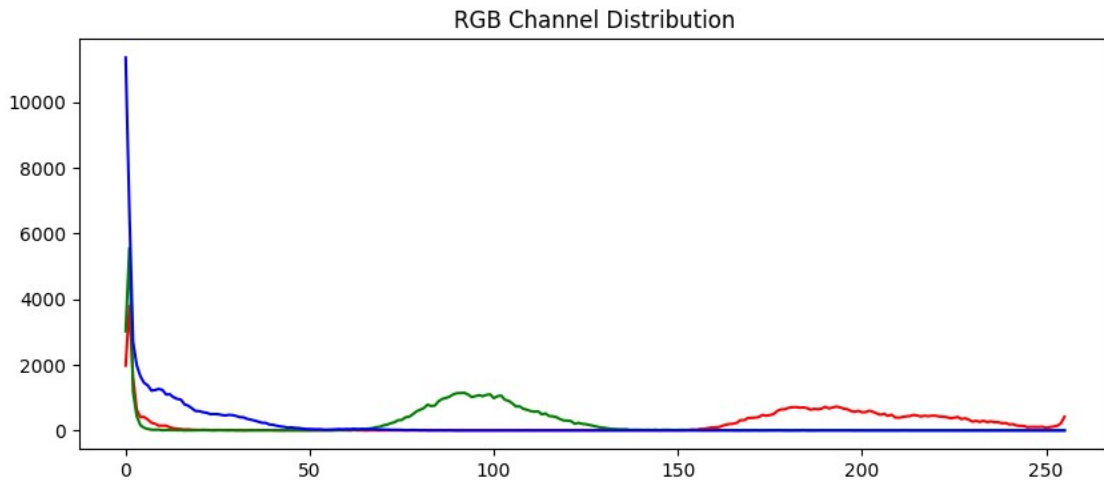


Figure : Results, Discussion and Evaluation.22. RGB Channel Distribution

5.1.5 Data Quality Assessment

To guarantee the integrity of the training process, a number of quality control checks were conducted:

a) Duplicate Detection: There were 128 duplicated images in the dataset.

b) Blur Detection: The mean score of blur was estimated at around 258.54 (assuming Laplacian variance). It means that the image sharpness of the dataset is rather high, which guarantees that the deep learning models will be able to extract fine-grained features such as microaneurysms.

c) Edge Detection: The skeleton of the retina was visualized using canny edge detection. This delineated the vascular tree and the limits of large exudates which proved that morphological characteristics needed to classify these were maintained.

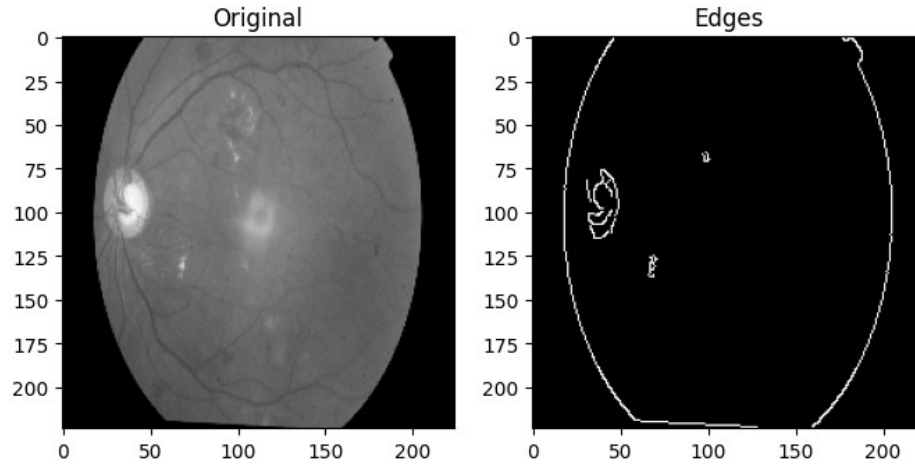


Figure : Results, Discussion and Evaluation.23. Edge Detection

5.2 Model Training Performance

Three state-of-the-art architectures, including VGG16, ResNet50 and InceptionV3, were trained in 30 epochs. The following observations were made regarding their learning trajectories.

5.2.1 VGG16 Training Performance

VGG16 showed gradual and smooth learning curve. The accuracy of the training started at around 20 percent and increased steadily to level around 70 percent. It is worth noting that the validation accuracy was a little bit higher than the training accuracy in the first stages indicating that dropout or strong weight initialisation was effectively used. The loss decreased smoothly from 1.9 to approximately 0.8. The convergence was smooth with little variations, which implied a well-tuned learning rate.

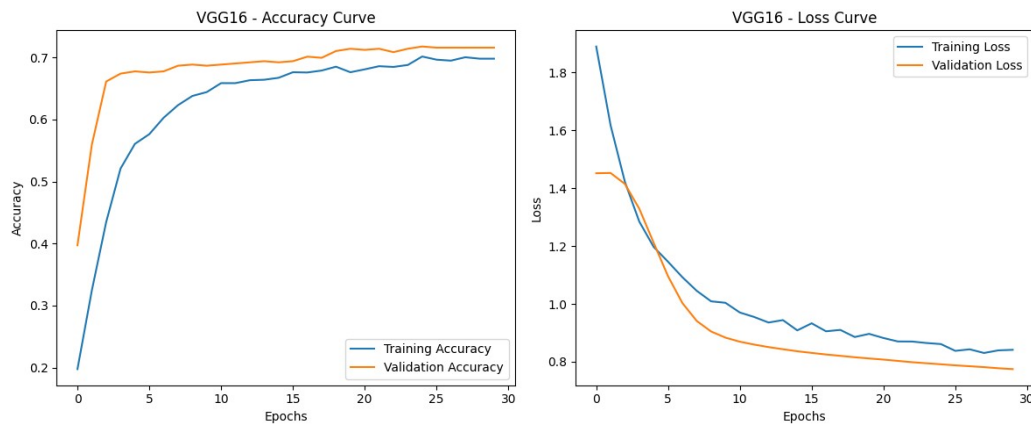


Figure : Results, Discussion and Evaluation.24. VGG16 Loss & Accuracy Curve

5.2.2 ResNet50 Training Performance

ResNet50 model took advantage of its residual connection to realize a quicker initial convergence as compared to VGG16. The model reached an accuracy of 70% within the first 10 epochs. It has had a very narrow distance between training and validation accuracy, a good measure of good generalization. ResNet50 had the smoothest loss curve of the three models and the final training loss of the model was about 0.82. The staircase effect common in deep networks was not present here indicating stable gradient flow.

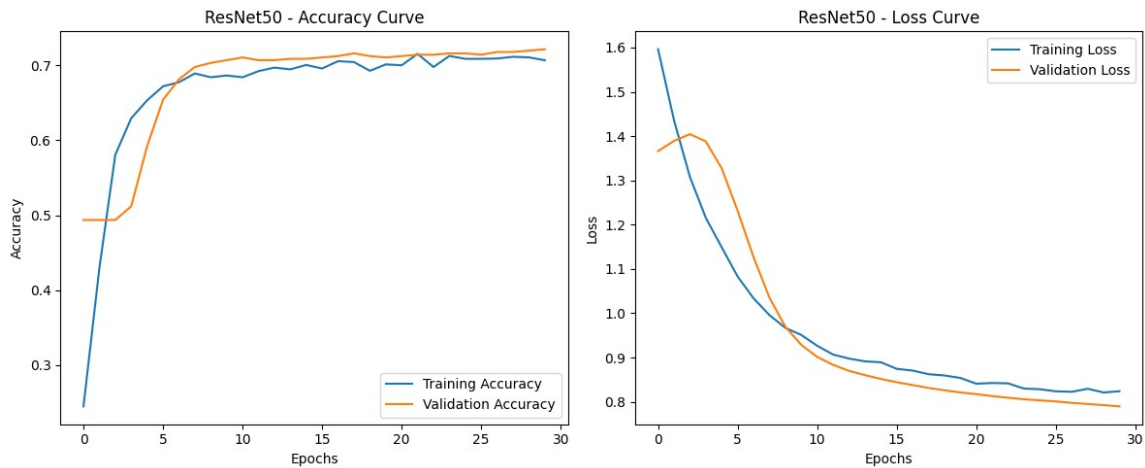


Figure : Results, Discussion and Evaluation.25. ResNet50 Loss & Accuracy Curve

5.2.3 InceptionV3 Training Performance

InceptionV3 had the most dynamic learning, which is probably because of its multi-scale inception modules. The model scored the highest total training accuracy, over 73. Nonetheless, the accuracy of validation exhibited more frequent variations than the ResNet50, especially in epochs 5-15. InceptionV3 was better at reducing loss although it was slightly volatile as it has a training loss of about 0.72. The fact that the training and validation accuracy were approaching each other towards the end of the 30 epochs indicates that the model was almost approaching its capacity in regards to the given dataset.

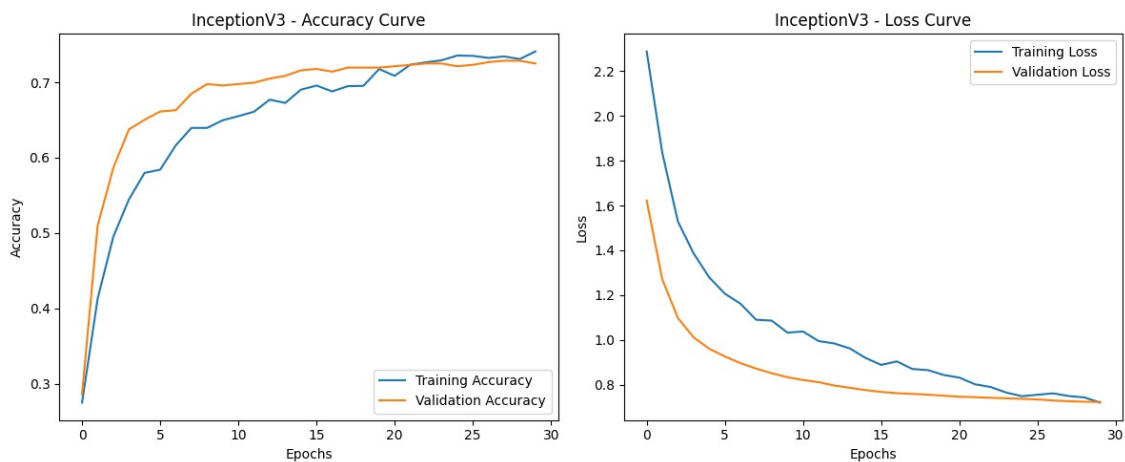


Figure : Results, Discussion and Evaluation.26. InceptionV3 Loss & Accuracy Curve

5.3 Model Evaluation Results

This section details the performance of the three architectures on the unseen test set (550 images), utilizing confusion matrices and classification reports to dissect their diagnostic capabilities.

5.3.1 VGG16 Evaluation

VGG16 had a total test accuracy of 73.64%. The model was exceptionally accurate in detecting healthy retinas (No_DR) and the precision and recall was 0.90 and 0.94 respectively. Nonetheless, it had serious problems with the minority classes. In the case of Proliferate_DR category, the recall was only a 0.05 meaning that the model failed to recognize 95% of the cases during this critical stage.

VGG16 Accuracy: 0.7364				
Classification Report:				
	precision	recall	f1-score	support
Mild	0.50	0.36	0.42	56
Moderate	0.58	0.83	0.68	150
No_DR	0.90	0.94	0.92	271
Proliferate_DR	0.40	0.05	0.08	44
Severe	0.67	0.14	0.23	29
accuracy			0.74	550
macro avg	0.61	0.46	0.47	550
weighted avg	0.72	0.74	0.70	550

Figure : Results, Discussion and Evaluation.27. VGG16 Evaluation Merics

There is a high diagonal in the case of "No_DR" (254 correct) and moderate (125 correct). Nevertheless, there were 31 false identifications of the Proliferate_DR images as the moderate ones, which indicates that VGG16 is unable to discriminate the fine-grained pathological characteristics between moderate and advanced stages of diseases.

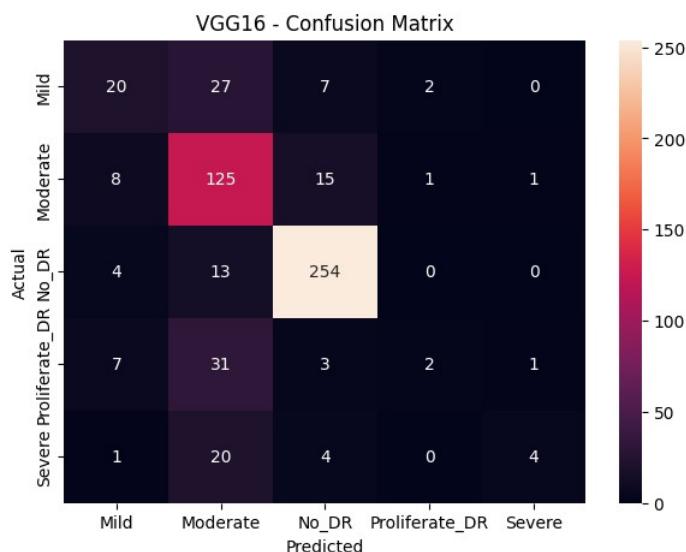


Figure : Results, Discussion and Evaluation.28. VGG16 Confusion Matrix

5.3.2 ResNet50 Evaluation

ResNet50 produced an accuracy of 73.27 in the test, which is a bit less than VGG16. Although the model was still sound in the case of the No_DR class (F1-score of 0.91), there was an overall failure in the minority classes. The accuracy, recall, and F1-score of both Proliferate_DR and Severe were 0.00.

```
ResNet50 Accuracy: 0.7327

Classification Report:

              precision    recall  f1-score   support

   Mild           0.51       0.36       0.42         56
  Moderate        0.59       0.81       0.68        150
   No_DR          0.86       0.96       0.91        271
Proliferate_DR    0.00       0.00       0.00         44
   Severe         0.00       0.00       0.00         29

 accuracy          0.73         0.73         0.73        550
  macro avg        0.39         0.43         0.40        550
 weighted avg      0.64         0.73         0.68        550
```

Figure : Results, Discussion and Evaluation.29. Resnet50 Evaluation Merics

The confusion matrix demonstrates that ResNet50 was unable to predict one case of either Proliferate_DR or Severe correctly. Rather it was skewed toward the "Moderate" and the No_DR class. This implies that even with its rich architecture, residual connections in the model were not sufficient to counter the severe class imbalance unless there was special weighting or oversampling.

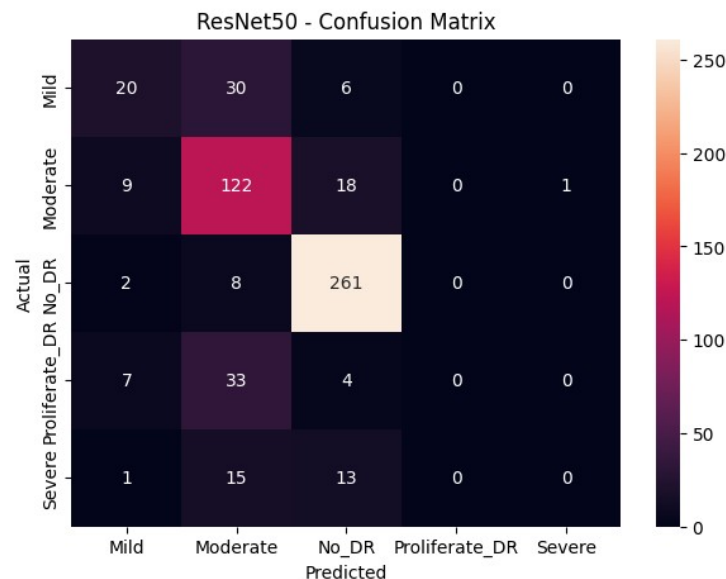


Figure : Results, Discussion and Evaluation.30. Resnet50 Confusion Matrix

5.3.3 InceptionV3 Evaluation

InceptionV3 was the best architecture with a test-accuracy of 76.36. InceptionV3, in contrast to the other models, showed the possibility to recognize samples of all

classes. It had the best Macro Average F1-score (0.53) and Weighted Average F1-score (0.74).

```
InceptionV3 Accuracy: 0.7636
```

Classification Report:

	precision	recall	f1-score	support
Mild	0.60	0.38	0.46	56
Moderate	0.62	0.81	0.70	150
No_DR	0.91	0.97	0.94	271
Proliferate_DR	0.46	0.25	0.32	44
Severe	0.44	0.14	0.21	29
accuracy			0.76	550
macro avg	0.61	0.51	0.53	550
weighted avg	0.74	0.76	0.74	550

Figure : Results, Discussion and Evaluation.31. InceptionV3 Evaluation Metrics

It has the strength of being multi-scale feature extraction, which enables it to attain a recall of 0.25 to Proliferate_DR and 0.14 to Severe--modest, though much better than ResNet50. The error of classifying the cases of advanced DR stages as the "Moderate" one is its main weakness since there are 28 cases of Proliferate_DR that are incorrectly classified in the category of the Moderate one.

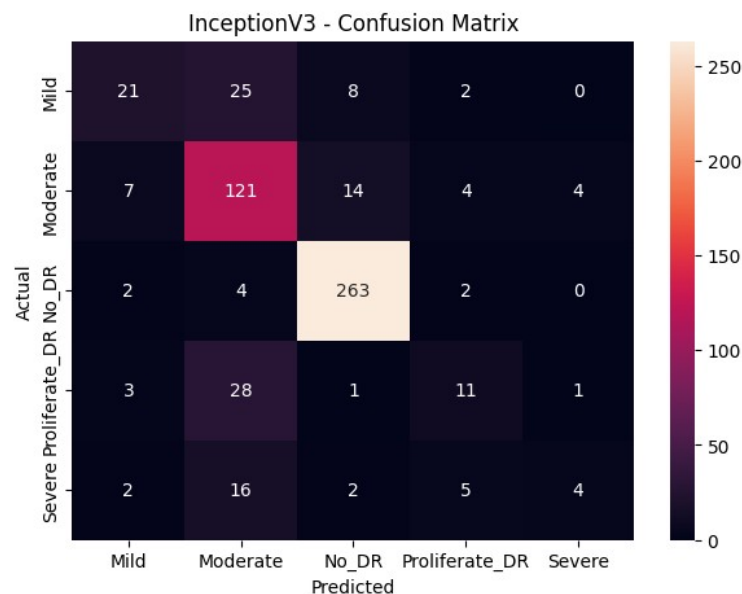


Figure : Results, Discussion and Evaluation.32. InceptionV3 Confusion Matrix

5.4 Comparative Analysis of Models

This section provides a side-by-side comparison of the three implemented architectures to identify the most suitable model for automated diabetic retinopathy (DR) diagnosis.

5.4.1 Summary of Performance Metrics

Model Architecture	Test Accuracy	Precision (Weighted)	Recall (Weighted)	F1-Score (Weighted)	F1-Score (Macro Avg)
VGG16	73.64%	0.72	0.74	0.70	0.47
ResNet50	73.27%	0.64	0.73	0.68	0.40
Inceptionv3	76.36%	0.74	0.76	0.74	0.53

Table : Results, Discussion and Evaluation.2. Performance Metrics of Models

5.4.2 Deep Dive into Architectural Performance

The performance variation between these models can be attributed to their internal structural design. The most successful one is InceptionV3, which had the highest accuracy and macro F1-score, indicating a good ability to deal with the multi-class nature of diabetic retinopathy. Its Inception modules make use of parallel convolutions of varying sizes, enabling the model to detect small microaneurysms and larger patches of exudates at the same time. This multi-scale model is probably the reason why it is the only model that attained non-zero recall of all five classes. The baseline standard VGG16 was the best performing with its outdated and simplistic sequential architecture. This implies that the 3x3 filters stacked deep in VGG16 gave a good feature extraction baseline, especially between the cases of No DR and diseased when using the given size of the dataset and the resolution of the image (224×224).

In contrast, ResNet50 showed the lowest performance in terms of macro F1-score. Even with residual connections, which generally serve to guard against vanishing gradients in deep networks, the model was highly biased towards majority classes like “No DR” and “Moderate” meaning that without methods such as class weighting, deeper networks might simply converge to majority class properties in skewed medical data.

5.4.3 Minority Class Sensitivity Analysis

The key comparison to be made when comparing the medical diagnosis models is the performance on high severity cases like Severe and Proliferative DR. InceptionV3 showed the most clinical potential because it was able to correctly identify some Proliferative DR cases with a recall of 0.25, whereas ResNet50 failed in this regard with a recall of 0.00. This emphasizes the relevance of model ability in identifying critical circumstances. The three models were also highly accurate in the class of No DR, above 0.86, and can be used in automated screening to

determine healthy patients. But the difference between No DR recall and Severe DR recall is the technical challenge that is mainly facing this pipeline.

5.4.4 Computational Efficiency vs. Accuracy

Although InceptionV3 is the most accurate player, it is much computationally more complicated than VGG16. But in a real-time image processing system, the milliseconds per image difference in inference time is insignificant on current equipment. Thus, the model that is chosen to be used in the proposed diagnosis system is InceptionV3, which has a greater capacity to identify diseased states and a lower computational cost when compared to the slight increase in computational cost.

5.5 System Deployment and Real-World Output Results

Following the comparative analysis, the model was integrated into the mobile application to validate its performance in a simulated clinical environment. This section presents the results of the system's end-to-end functionality, from user registration to automated diagnostic reporting. Fundus Eye is a cross-platform mobile healthcare application developed using Flutter, designed to assist in the early screening of Diabetic Retinopathy (DR) through AI-based analysis of retinal fundus images. The system integrates on-device machine learning (TensorFlow Lite) with a secure cloud backend (Firebase) to provide a complete solution for patient data management, image acquisition, automated diagnosis, and report generation.

5.5.1 User Authentication and Provider Access

The application ensures secure access for healthcare providers to maintain data privacy and clinical accountability. The system provides a streamlined sign-up and login process for clinicians. User credentials are authenticated through the Firebase backend, ensuring that only authorized users can access patient data and diagnostic reports.


FundusEye
Now Funduscopy at Your Hand

Sign up

Please enter log in details below

Name

Shafiq Ahmed khan

Email

Shafiq.ahmed@gmail.co.

Password

.....

☒ I agree with [privacy](#) and [policy](#)

sign up

Figure : Results, Discussion and Evaluation.33 User Authentication Interface

5.5.2 Patient Enrollment and Data Entry

To maintain a structured clinical workflow, the system captures essential patient information before diagnostic analysis. Healthcare providers can create patient profiles by entering key demographic details such as name, age, and gender. This information is stored in real-time within the database, allowing diagnostic results generated by the model to be linked with individual patient records for future reference and longitudinal tracking.


FundusEye
Now Funduscopy at Your Hand

Add Patient

Ahmed

20

Male

Pick Eye Image

Submit

+ Add Patient

Figure : Results, Discussion and Evaluation.34 Patient Enrollment and Data Entry Form

5.5.3 Flexible Image Acquisition

Clinicians can either capture retinal images directly using the device camera or upload existing images from the device gallery. Once an image is selected, it is processed using TensorFlow Lite on the device, ensuring fast and efficient diagnostic inference without heavy reliance on external servers.

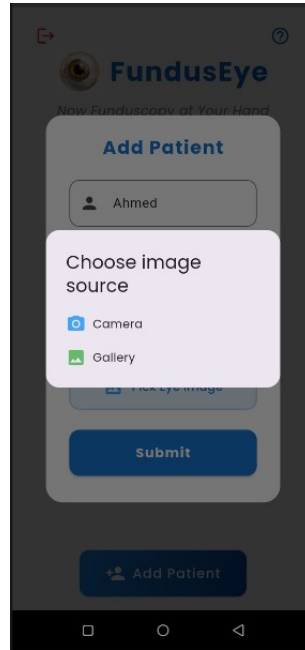


Figure : Results, Discussion and Evaluation.35 Image Selection Interface

5.5.4 Clinical Dashboard and Case Management

A centralized dashboard is provided to help clinicians efficiently manage patient data and review diagnostic history. The dashboard displays a list of registered patients along with their basic details. It allows healthcare providers to quickly access patient records, review previous diagnoses, and manage multiple cases simultaneously, improving overall workflow efficiency.

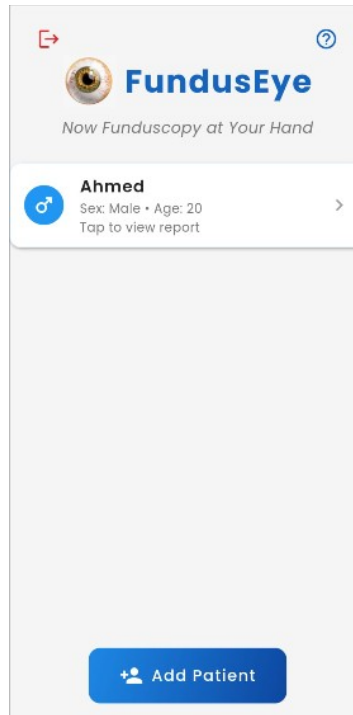


Figure : Results, Discussion and Evaluation.36 Patient Management Dashboard

5.5.5 Automated Diagnosis and Medical Guidance

The system generates a comprehensive diagnostic report based on the analysis of retinal images. The application classifies the severity of Diabetic Retinopathy and presents the results in a clear and understandable format. In addition to the diagnosis, the system provides automated medical advice to support clinical decision-making, such as recommending immediate specialist consultation for severe cases. The application also includes reporting features that allow users to share or print diagnostic reports, facilitating communication between primary healthcare providers and ophthalmology specialists.

Patient Report

Name: Ahmed

Age: 20

Gender: Male

Patient ID: dTFa2YrIG2rOYX9SJ9Q1

Condition: Severe Diabetic Retinopathy

Medical Advice

Seek specialist attention immediately.

Note: This is AI generated report. Please consult it with your doctor.



Share



Print

Figure : Results, Discussion and Evaluation.37 Automated Diagnostic Report with Severity Classification

5.5.6 Use Case Diagram of System

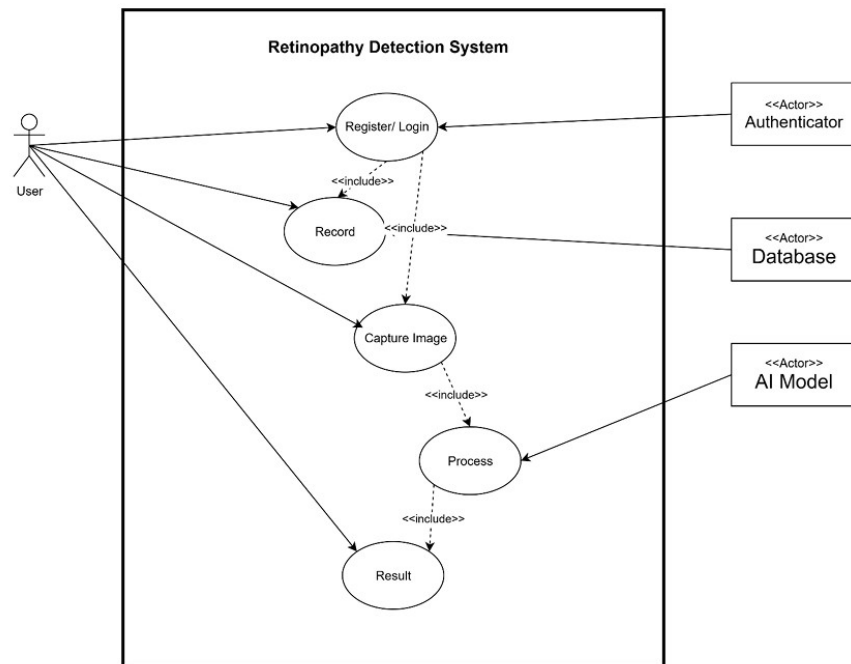


Figure : Results, Discussion and Evaluation.38 Use Case Diagram (self)

5.5.7 Sequence Diagram of System

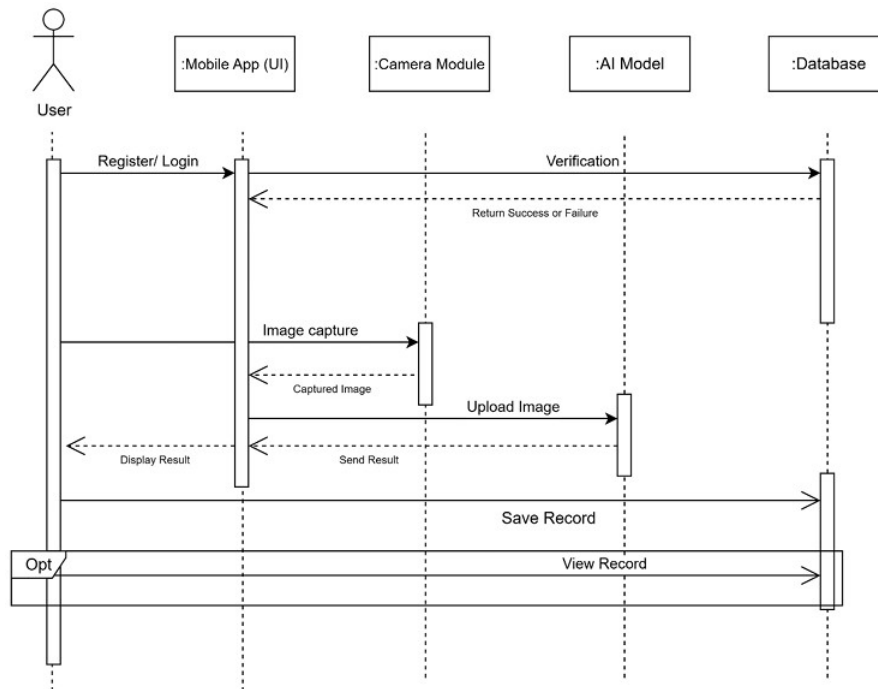


Figure : Results, Discussion and Evaluation.39 Sequence Diagram (self)

5.5.8 Activity Diagram

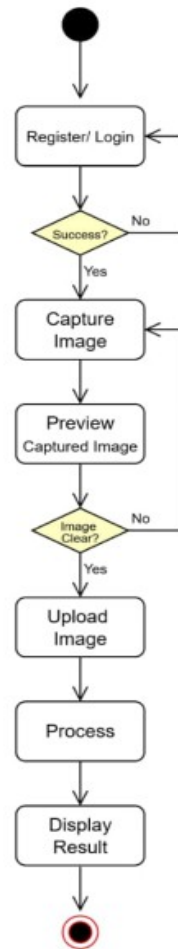


Figure : Results, Discussion and Evaluation.40 Activity Diagram

5.6 Discussion of Results

5.6.1 Key Findings from EDA and Models

The study highlights a direct correlation between data distribution and model success. The 97% recall for the "No_DR" class across all models confirms that deep learning is highly effective for primary screening (distinguishing healthy from diseased eyes). However, the performance drop in "Severe" and "Proliferate" classes is a direct result of the class imbalance identified during EDA (where these classes comprised less than 10% of the total data).

5.6.2 Model Strengths and Limitations

The models demonstrate high specificity for normal cases, making them valuable for reducing the workload of ophthalmologists by filtering out healthy patients. The primary limitation is "gradient toward the majority." In medical contexts, missing a "Severe" case (False Negative) is more dangerous than misclassifying a "Mild" one. The current models show a high rate of False Negatives for advanced disease stages.

5.7 Summary of Results

The experimental results conclude that:

- InceptionV3 is the most effective model for automated diabetic retinopathy classification.
- Data quality assessments (identifying 128 duplicates and ensuring a blur score of 258.54) were vital in ensuring the test results were not artificially inflated.
- The system excels at detecting the absence of disease but requires more diverse data to perfectly distinguish between the four stages of DR progression.
- The InceptionV3 model was successfully integrated into the FundusEye mobile application developed using Flutter, TensorFlow Lite, and Firebase to provide a functional end-to-end solution for patient management, real-time diagnosis, and clinical reporting.

Chapter 6 : Critical Evaluation and Limitations

This final chapter serves to synthesize the entire research project by reflecting on the developed deep learning pipeline for automated brain stroke classification. It first provides a comprehensive critical evaluation of the research design, methodology, and data constraints to demonstrate reflexivity and establish academic credibility. Following this critical assessment, the chapter concludes by summarizing the core findings, directly answering the initial research questions, and outlining actionable recommendations for future work and clinical implementations

6.1 Critical Evaluation

Reflecting on this study, developing a deep learning pipeline and embedding it into a mobile application to classify diabetic retinopathy has been an educative and fulfilling experience. This part is a reflection of the decisions that have been made concerning the research design, methodology, and problems that have arisen during the project. The most important outcomes of this work are the development of the so-called system that should support the disadvantaged groups of people providing them with a cost-efficient alternative to the more costly medical imaging solutions.

Technically, much attention was devoted to preprocessing data to provide a stable model and successful training. Methods like reducing the size of pictures to Resolution: 224x224 (keeping the same color representation) and normalizing the pixel values were significant in ensuring smooth convergence. The investigation of various pre-trained CNN architectures yielded insights on the best-performing model, where InceptionV3 was the highest performing model with a test accuracy of 76.36 and a macro F1-score of 0.53 because of the high multi-scale feature extraction ability. Comparatively, ResNet50 could not identify minority classes, which implies that more advanced models can be biased against dominant classes when trained on imbalanced datasets without an effective balancing mechanism, and VGG16 can be a good baseline, but it is more likely to issue false positives by considering severe cases as mild.

Class imbalance in the dataset was one of the biggest obstacles experienced in this research. The classes distribution was grossly lopsided with a high number of samples within the category of No DR and only a few samples in the severe stages. This imbalance influenced the model to learn and identify minority classes accurately, and gave biased predictions to majority classes.

6.2 Limitations

Although this study has positive results, there are a few limitations of the study. The most notable is the imbalance in the number of classes within APTOS 2019 dataset, where the No DR class has the most images (1,805), whereas the Severe one comprises only 193, which is only 5.2 percent of the dataset.

This disproportion resulted in the models giving preference to the majority classes resulting to the bias that adversely influenced the detection of late stages of the

disease. Particularly, the ResNet50 models had many false negatives in severe cases, which is a critical issue in medical practice. Also, the fact that one dataset is used restricts the strength of the model since it might not be consistent in different real-life aspects like lighting variations or image quality across different mobile devices.

Chapter 7 : Conclusion

This study has managed to show how a deep learning-based pipeline can be developed into a mobile application that can be used to screen diabetic retinopathy with high accuracy, especially in low-resource environments. The results give a definite picture of model performance, data issues and feasibility of deployment which directly answers the research questions set.

As per Research Question 1, InceptionV3 has attained the highest multi-class diagnostic accuracy followed by VGG16 whereas ResNet50 demonstrated relatively low performance. This substantiates the fact that architecture design has a great impact on the capability of performing detailed classification task like diabetic retinopathy.

Research Question 2 found that InceptionV3 was the most effective model at the classification of Severe and Proliferative DR with imbalance in classes because it had non-zero recall of these important classes. Conversely, ResNet50 did not identify such instances, which underscores the shortcomings of more intricate architectures in the presence of imbalanced data without the technique of handling such data.

Based on Research Question 3, despite the fact that InceptionV3 is computationally more complicated than VGG16, the distinction in real-time processing speed is small on current hardware. This means that better diagnostic performance can be attained without much loss in efficiency in mobile deployment.

Standardized preprocess methods, such as image resizing, normalization, and uniform color formatting, were also found to be crucial in ensuring consistent model training and consistent convergence across all architectures according to Research Question 4.

InceptionV3 provides the best balance between accuracy, minority classes detection and is deployable as per Research Question 5. Its incorporation into a secure, cloud-based mobile application indicates its appropriateness in terms of being a scalable and feasible real-time diabetic retinopathy diagnosis solution in resource-constrained settings.

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